Avian Reovirus Infections

家禽里奧病毒感染症

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introduction

- Avian reovirus (ARV) is ubiquitous in commercial poultry, especially in broilers
- Age-associated Resistance
- Avian reovirus was first isolated in 1954 from the chicken respiratory tract and later in 1957 from a case of synovitis(滑膜炎) in chickens
- The virus was identified in 1972 as a reovirus

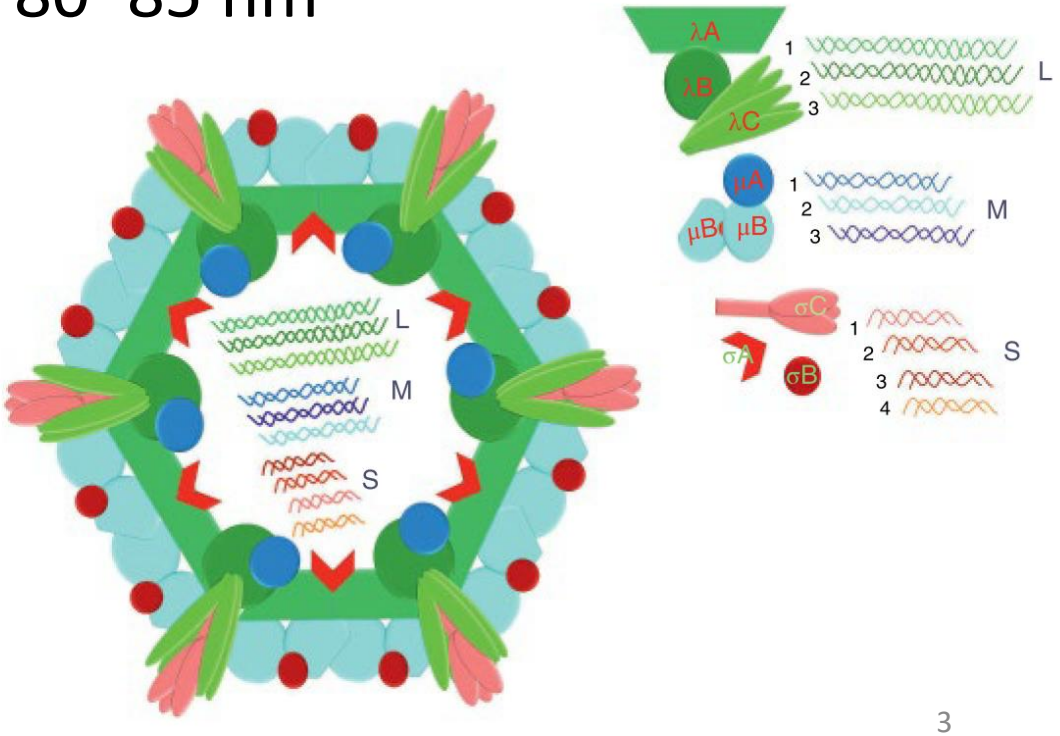
Avian reoviruses (ARV)

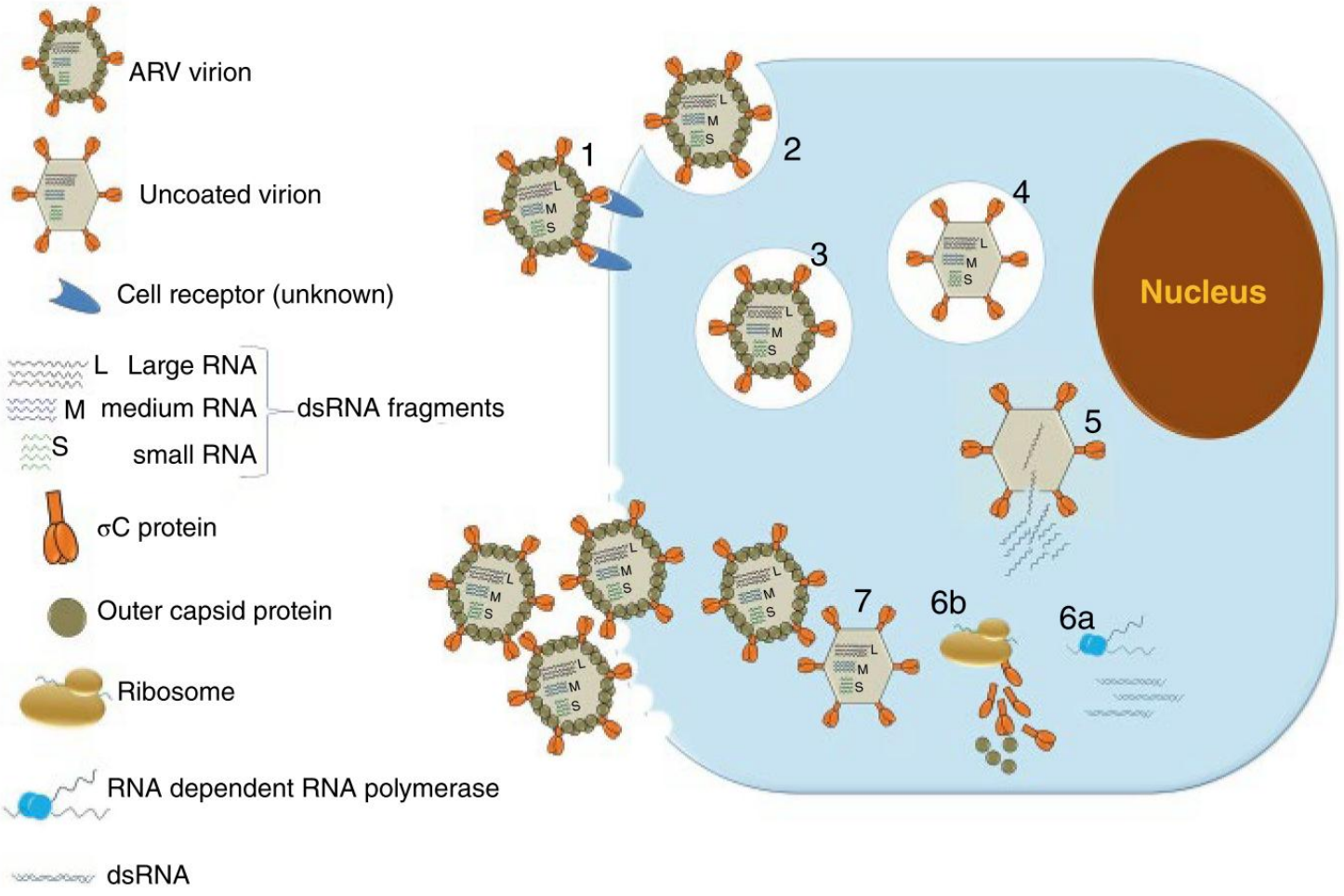
- Reoviridae family , genus *Orthoreovirus*
- nonenveloped, ds RNA
- icosahedral capsid, 80–85 nm
- 呼腸孤病毒

  Respiratory

  Enteric

  Orphan





virion exit followed by cell lysis.

TABLE 15.1 Characteristics of Reovirus Infections

Virus/Hosts/Serotypes	Disease/Symptoms	Transmission/Diagnostic Specimen
Orthoreovirus		
Reoviruses in mammals Serotypes 1–4	Asymptomatic infection, experimental disease	Fecal–oral route; systemic infection
Reoviruses in poultry Reoviruses in many bird species, multipleserotypes	Tenosynovitis/arthritis Respiratory disease, enteritis, with weight loss, stunted growth; often subclinical	Fecal–oral route; systemic infection Feces, serum, multiple target tissues
Orbivirus		
Bluetongue virus Sheep, cattle, goats, deer Serotypes 1–25 (27?)	Bluetongue Fever, hyperemia, cyanosis, edema, oral, cavity erosions, nasal discharge, lameness	Vector transmission: *Culicoides* spp. Blood—virus detection; serum—antibody testing; spleen, lung, lymph nodes
African horse sickness virus Horses; donkeys, mules, zebras (subclinical)Serotypes 1–9	Respiratory or cardiovascular failure Fever, edema	Vector transmission: *Culicoides* spp. Blood—virus detection; serum—antibody testing, spleen, lung, lymph nodes
Equine encephalosis viruses Horses; donkeys, mules, zebras (subclinical)Serotypes 1–7	Subclinical often Fever, African horse sickness-like disease	Vector transmission: *Culicoides* spp. Blood—virus detection; serum—antibody testing, spleen, lung, lymph nodes
Rotavirus		
Rotaviruses Virtually all species	Gastroenteritis/diarrhea	Fecal–oral route Feces

Avian reoviruses (ARV)

- 耐熱 (60°C for 8-10 hrs)
- pH3

- ubiquitous among domestic poultry
- replicates in the gut of avian species
- pathogenic strains can affect tendon
- in broilers and turkeys:

  arthritis/tenosynovitis（關節炎/腱鞘炎）
  - Swelling of the hock joints (足關節)
  - lesions in the gastrocnemius tendons(腓腸腱)

Viral arthritis 病毒性關節炎

- 早期稱tenosynovitis (腱鞘炎)
- 由ARV感染造成，常與*Mycoplasma synoviae* or *Staphylococcus spp.* 混合感染
- co-infection: 球蟲、IBD
- mainly affects meat-type chickens

Host

- chickens, turkeys
- age-associated resistance
  - 1日齡最敏感
  - 自然感染多4~7週齡雞 →經濟損失
  - 10週齡以上不敏感
- is commonly found in the digestive and respiratory tracts of clinically normal chickens and turkeys →difficult to interpret the relationship between virus isolation and disease

Transmission

- horizontal and vertical routes
- horizontal:

  粪便→ 口/呼吸道→ 消化道
- Vertical: 介蛋（非常少）

Incubation period

- depending upon the virus pathotype, age of host, and route of exposure
- 2 wks chickens→

  1 day: footpad inoculation

  9 day: intratracheal inoculation

  11 days: intramuscular, intravenous inoculation

  13 days: contact exposure

Clinical Signs

- 行動不便、lameness (跛行)
  - usually appears at 4–5 weeks of age although the infection may have occurred soon after hatching
- stunted (發育不良)、生長慢
- 羽毛生長不良
- 產蛋下降10%
- 死亡率不高 (5%)
- 發生率最高可達100%

- 經濟損失:
  - 非由死亡率導致
  - diminished weight gain 體重減輕
  - poor feed conversion 飼料換肉率差
  - reduced marketability 適銷性降低



Stunting syndrome in broilers


aaaP


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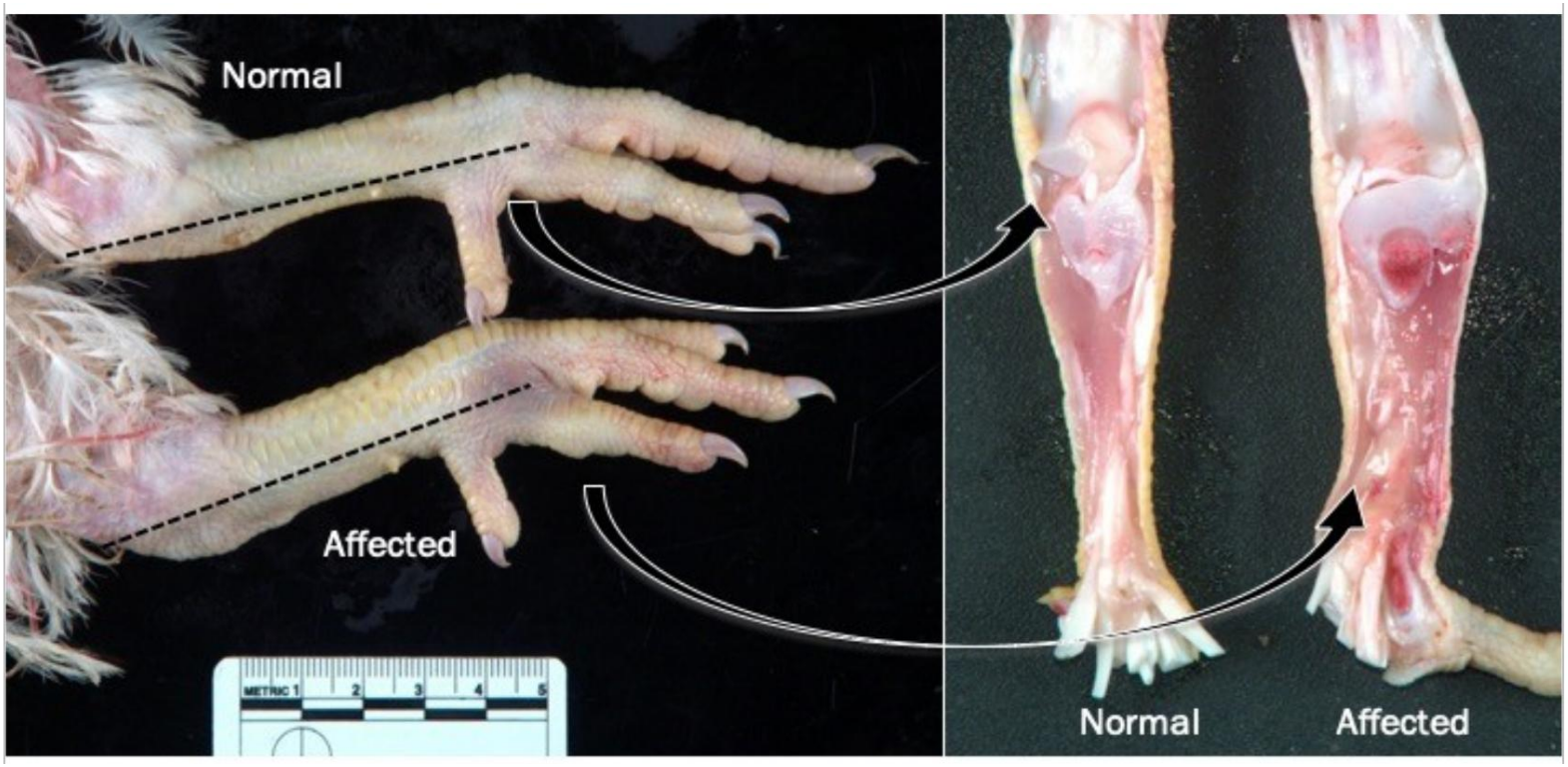
Gross

- swelling hock joints, 切開後有淡黃色或血樣滲出物





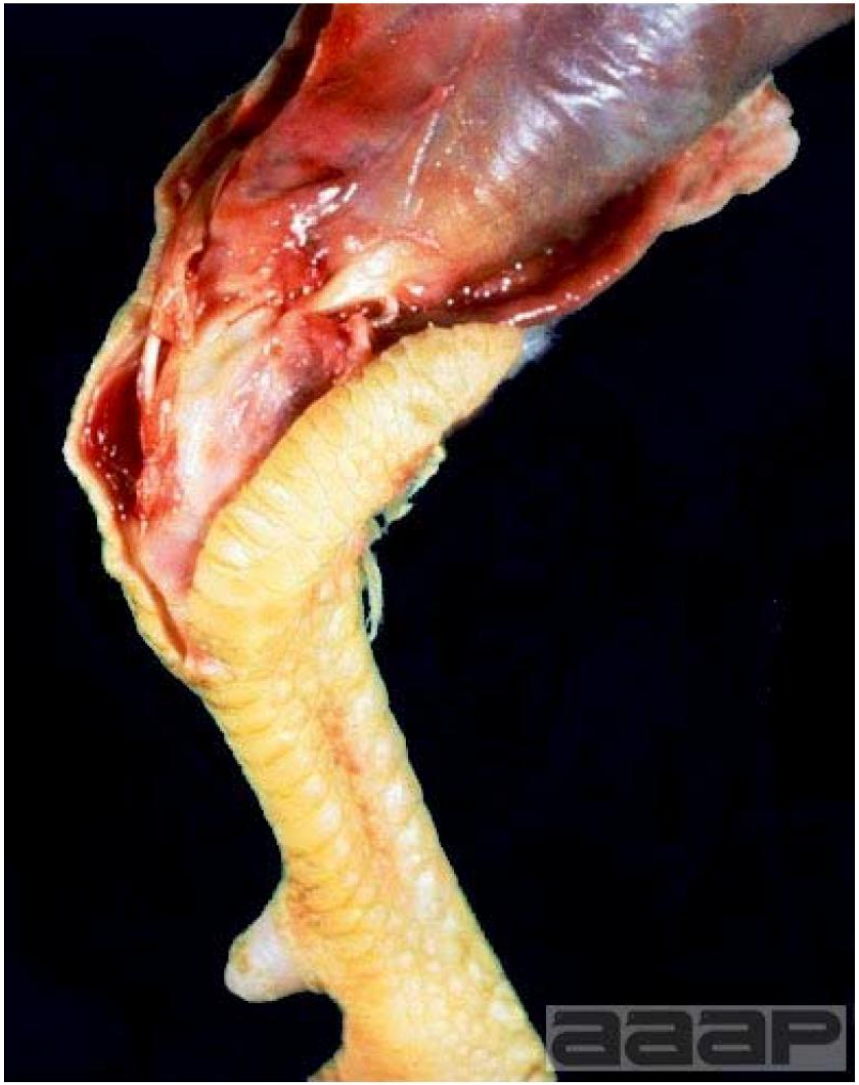

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Courtesy of Dr. Jenny Nicholds.


(B)


(C)





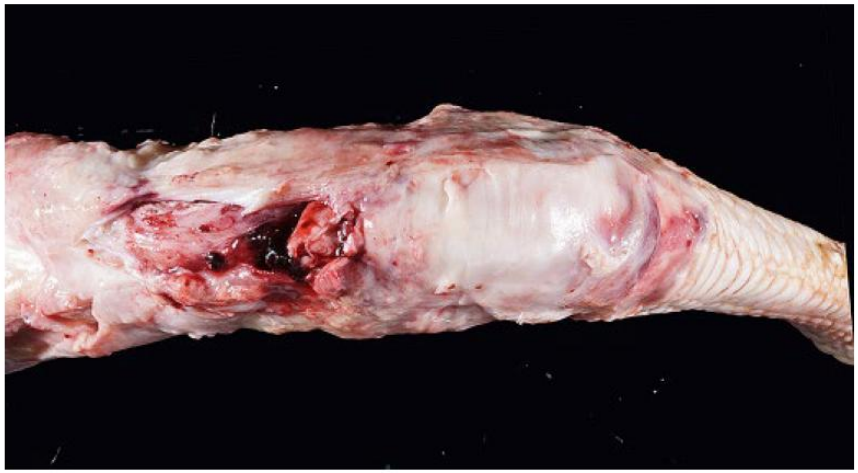




Photo from MSD




1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
CM


(A)


(B)

Gross

- 足關節病變最清楚，剪開可見關節液增加
  - 清澈透明→ ARV
  - 混濁 → 細菌性感染

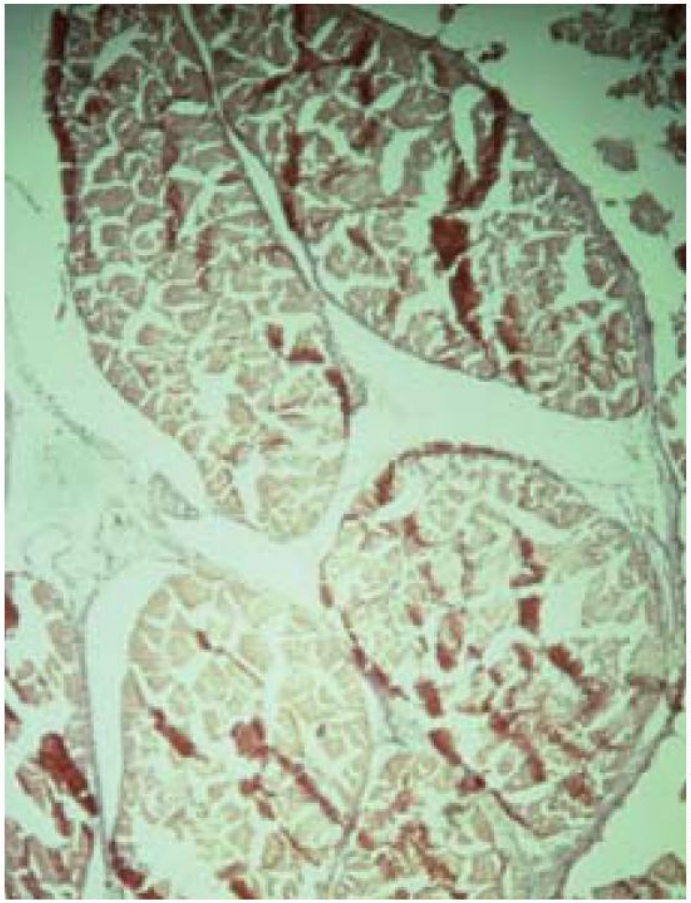
- 腺胃出血、大小不一





組織病變

- 滑膜下層可見漿細胞及淋巴球浸潤
- 水腫、壞死



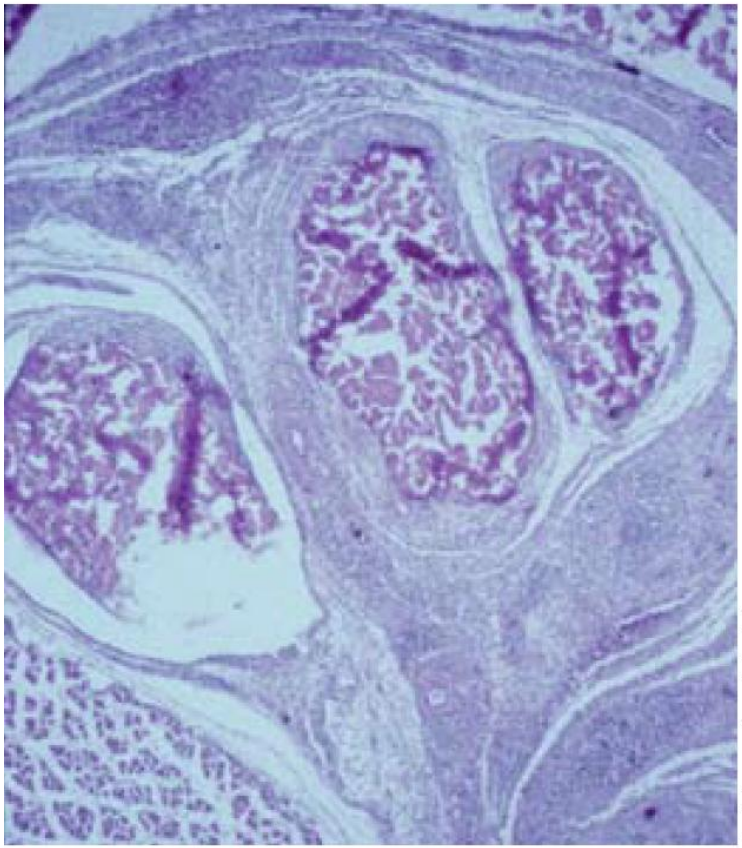
L Van der Heide

Fig.27.11: Cross-section of normal metatarsal flexor tendons.



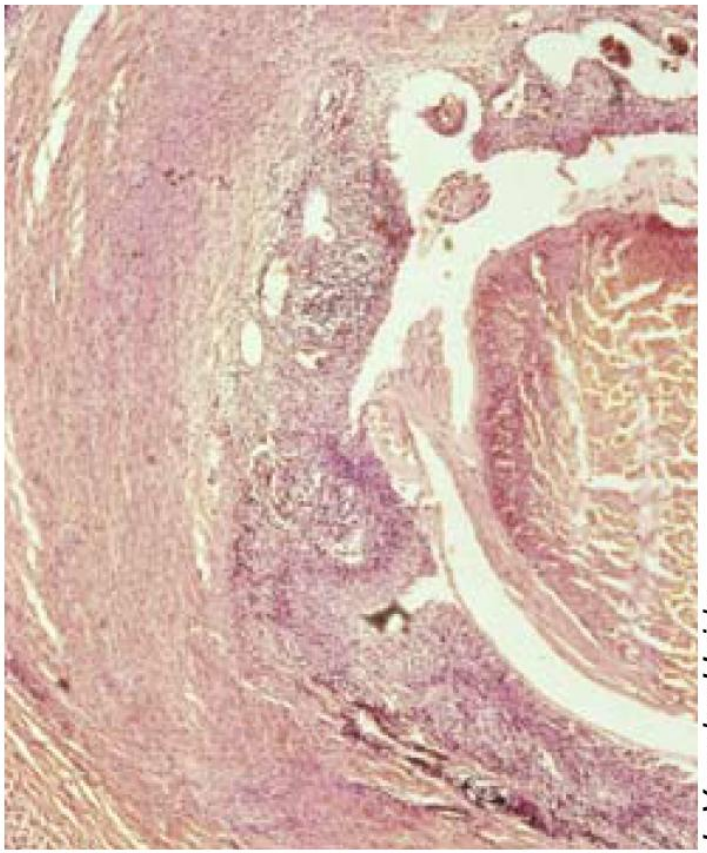
L Van der Heide

Fig.27.12: Cross-section of teno-synovitis infected flexor tendons: tendon sheaths have infiltration of lymphocytes primarily.



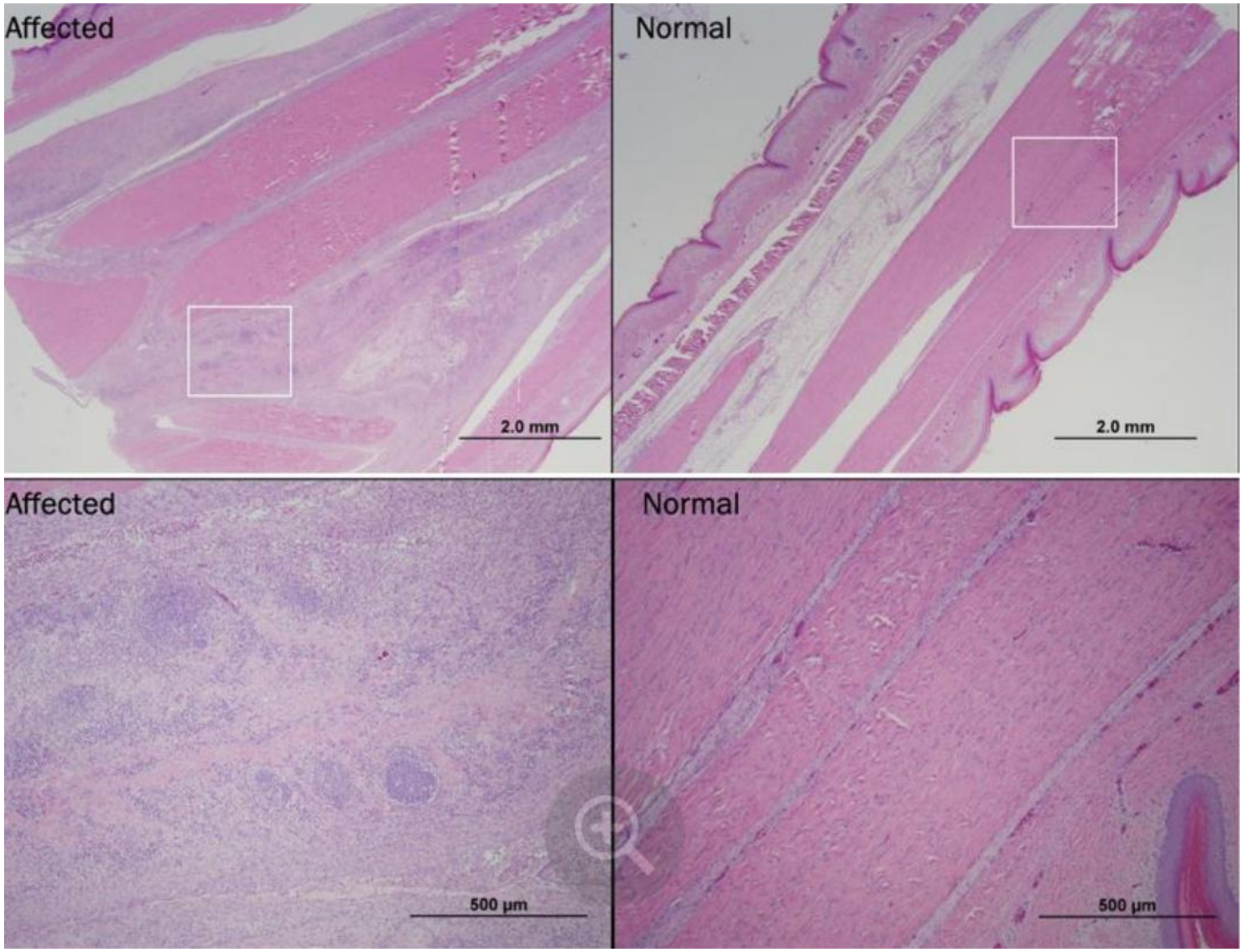
L Van der Heide

Fig.27.13: Cross-section of more advanced tenosynovitis infected flexor tendons: granulomatous tissue and beginning fibrosis.



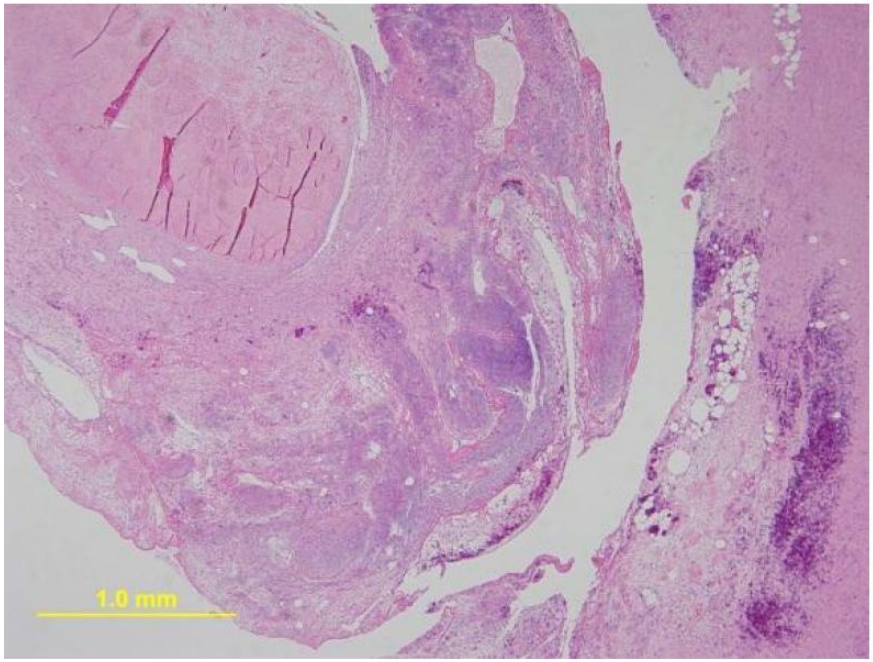
L Van der Heide

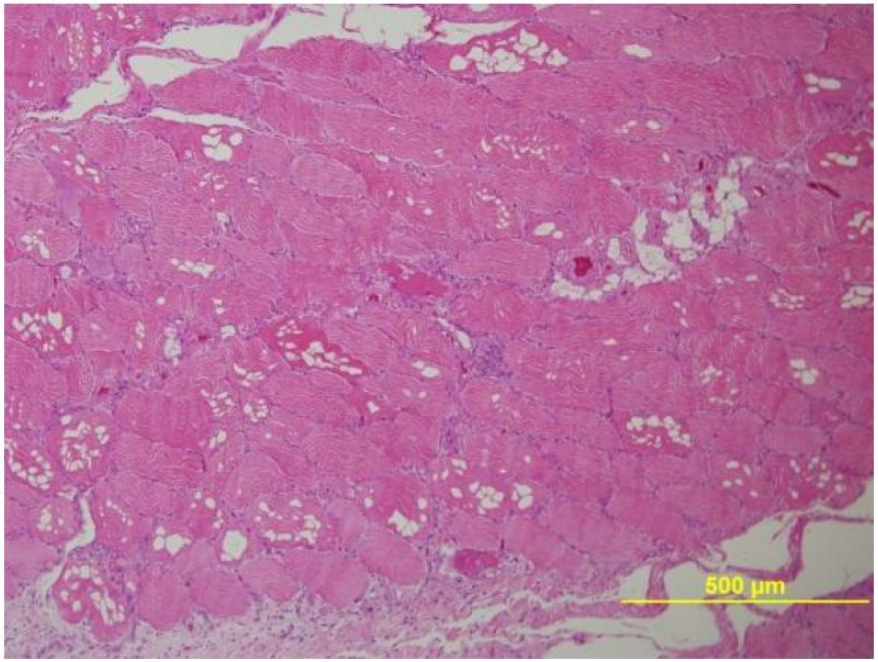
Fig.27.14: Cross-section of chronic tenosynovitis: marked fibrosis with some mononuclear cells.



Histopathology of normal and affected tendons in poultry with viral arthritis..

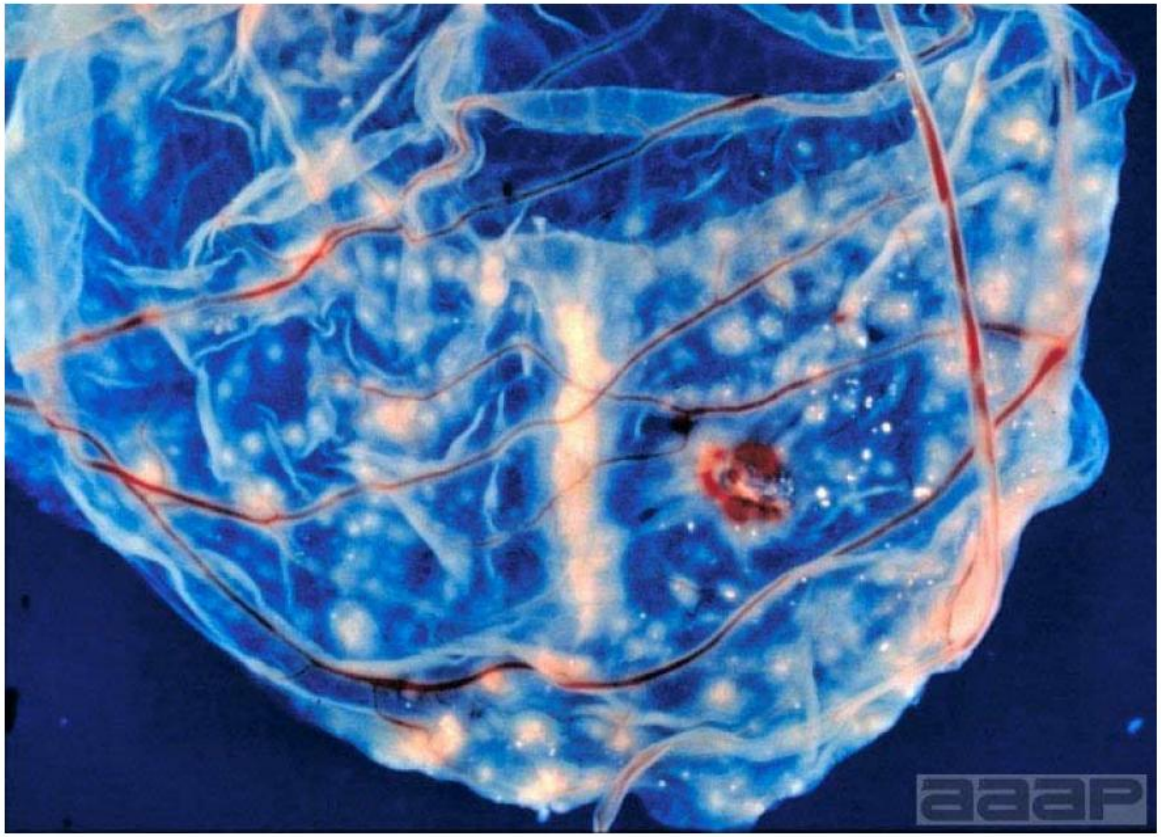
Courtesy of Dr. Jenny Nicholds.


1.0 mm

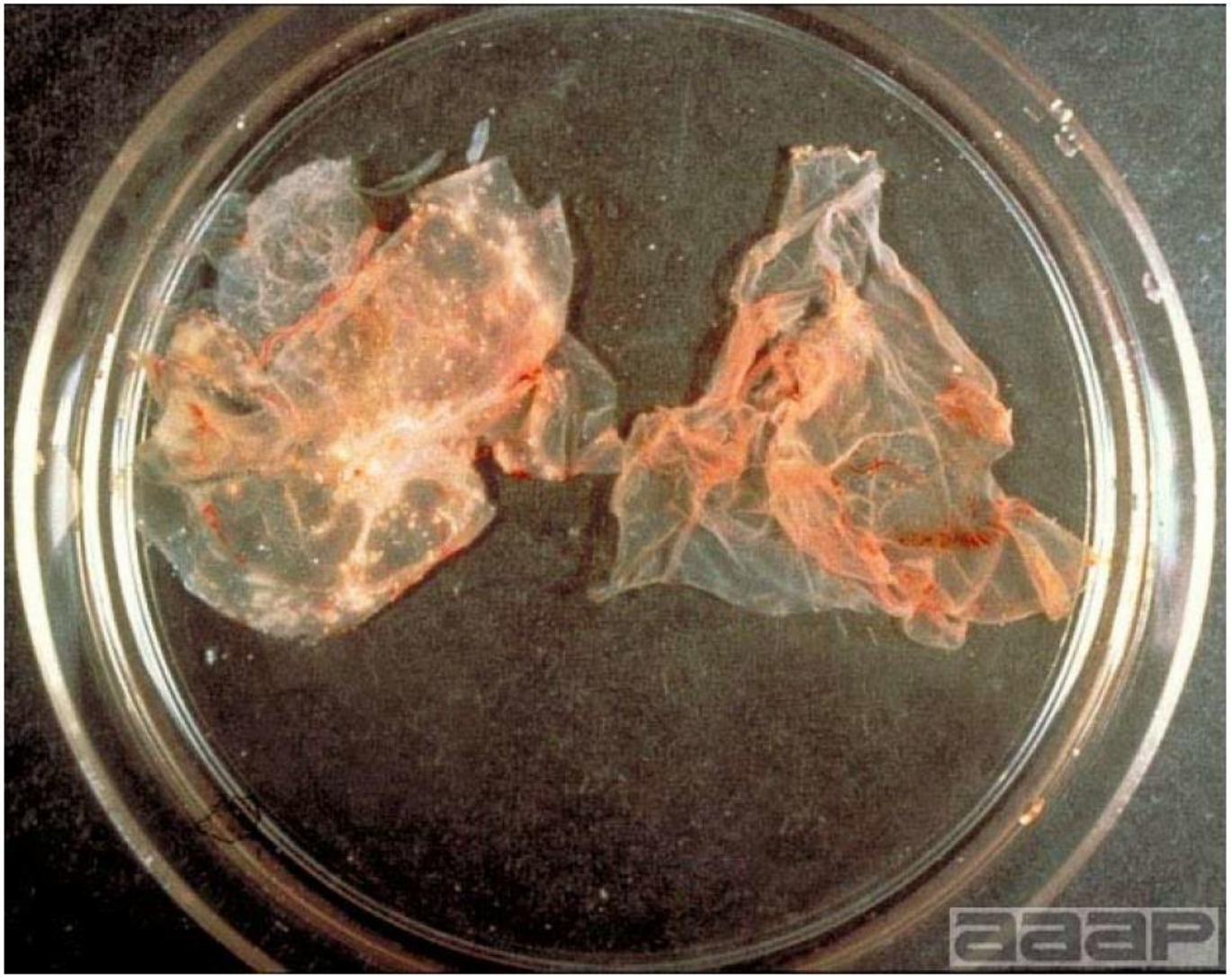

500 µm

Diagnostic methods

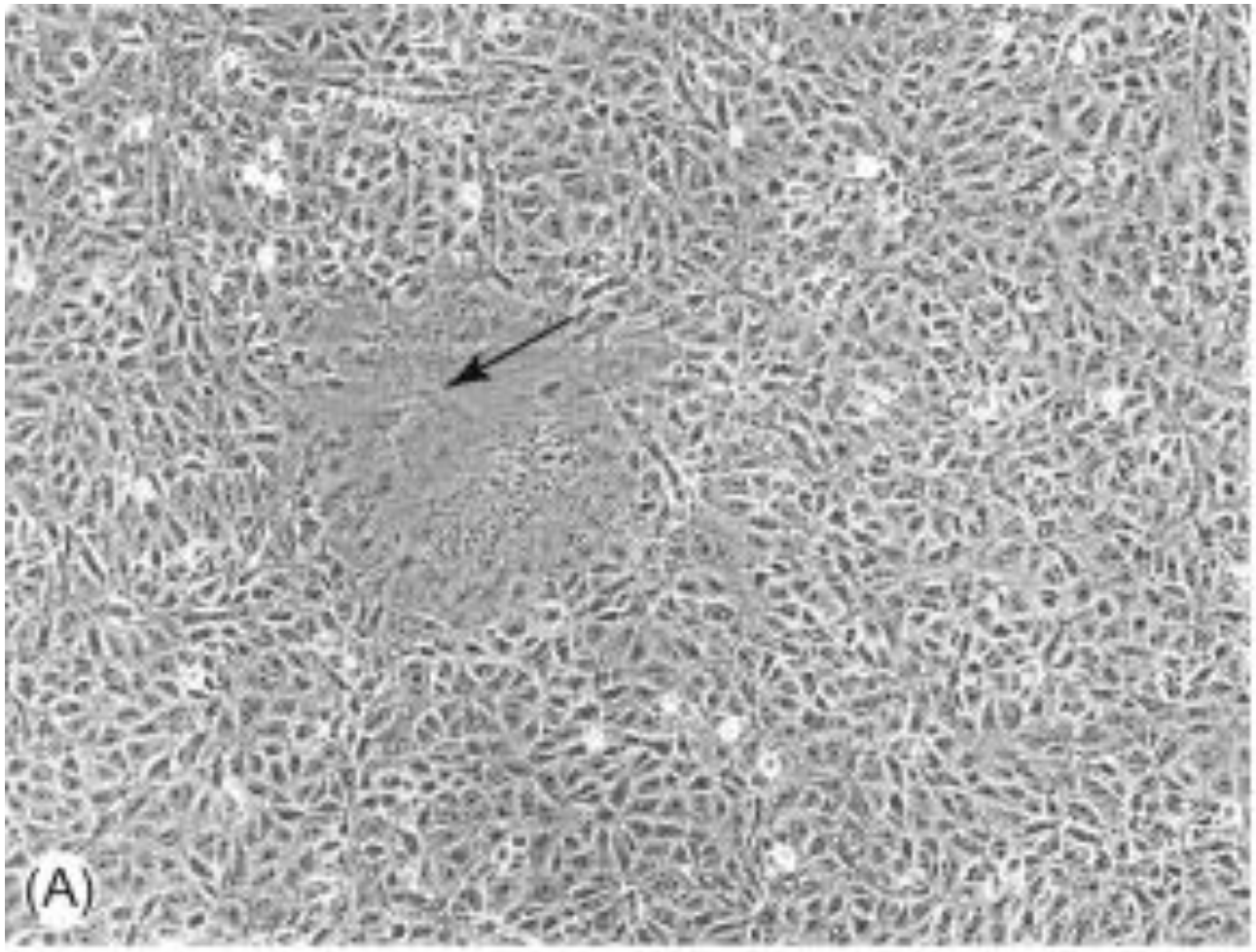
- Virus isolation
  - 雞胚胎，CAM接種→接種後約7-8日死亡
  - Vero, BHK-21, PK-15
- electron microscopy
- immunofluorescence
- agar gel precipitin test (AGP)
- RT-PCR
- ELISA: 看抗體力價均勻度



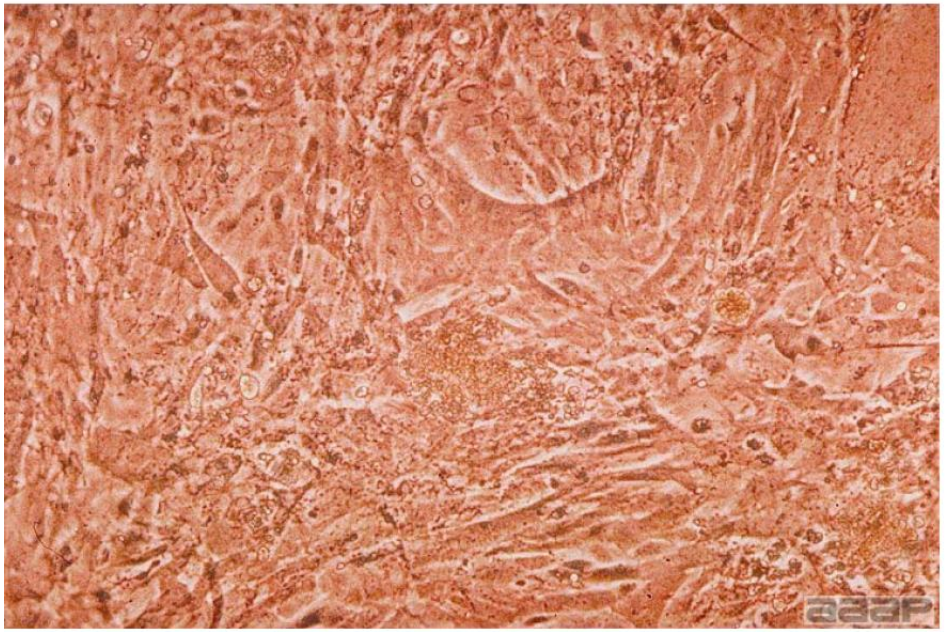
SLIDE 24. Inoculation of 10-day-old embryonating chicken eggs on the dropped chorioallantoic membrane (CAM) results in the development of whitish plaques in 3 to 5 days post-inoculation.

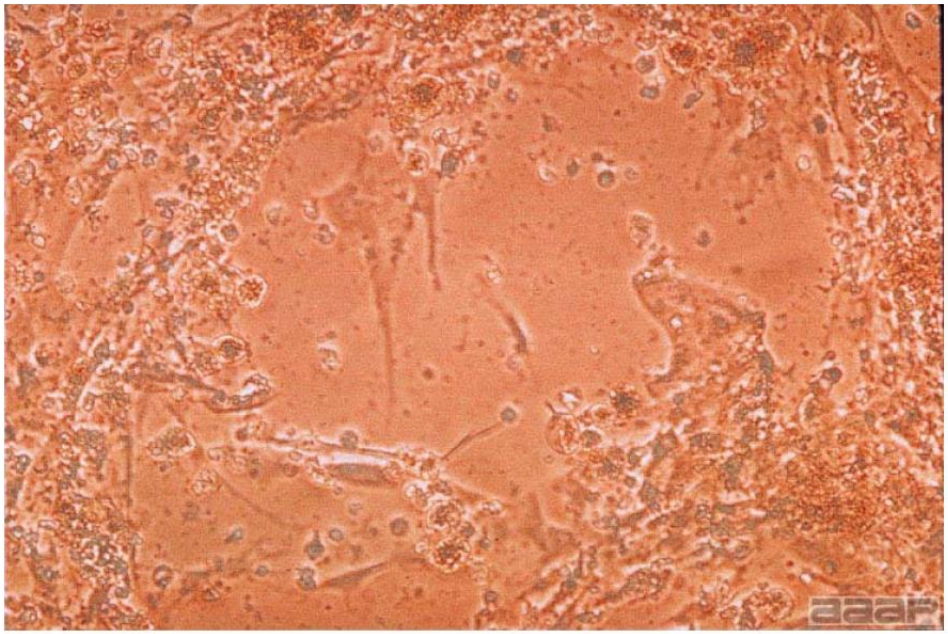



aaap



Avian reovirus in Vero cells





CK cell



Prevention and Control

- Biosecurity

  ARV對環境抵抗力強

  因為會從消化道長期排毒，得病→淘汰或隔離

  空欄

  碘

Vaccination

- 有活毒跟死毒疫苗，兩者並用效果較好
- 目前對疫苗接種計畫尚無共識
- 母雞開產前2-3週注射疫苗
  - Live vaccines should be administered only prior to the onset of egg production in order to prevent trans-ovarian transmission of the vaccine virus
- 以移行抗體保護小雞

  (可保護小雞在孵出2-3週不被感染)